A Diagnostic Study of Multi-Agent LLMs for Real-World Debates

Anonymous Authors[1]

Abstract

Multi-agent LLM debates are increasingly deployed in domains such as policy analysis and city planning, where no objective ground truth exists. Despite this, debate quality is typically evaluated using outcome-based proxies such as LLM-as-judge scores that provide little insight into whether meaningful deliberation has occurred. Additionally, consensus and majority vote are viewed as ideal goals without analyzing the underlying interaction dynamics beneath them. In this work, we introduce a diagnostic evaluation framework that measures debate quality by measuring both the outcome and the process. Grounded in deliberative theory, our framework defines four interpretable process-level metrics capturing engagement, responsiveness, influence asymmetry, and balance, and two outcome-based metrics capturing stability and agent utility.

Across both objective benchmarks and real-world domains, we find that process-level diagnostics are consistently more informative than commonly used outcome-based proxies. They better reflect correctness when ground truth exists and align more closely with human judgments of deliberative quality when it does not, revealing interaction failures that outcome-only measures fail to capture. These results demonstrate that process-level diagnostics are necessary for reliable evaluation of multi-agent debates and provide a principled foundation for analyzing and designing deliberative LLM systems.

1. Introduction

Large Language Models (LLMs) are increasingly used in multi-agent settings to simulate debate, discussion, planning, and collective reasoning. In domains with clear ground truth, such as question answering, mathematics, and problem solving, prior work has shown that interactions among multiple agents can improve factual accuracy, reduce hallucinations, and encourage diverse reasoning paths (Liang et al., 2024; Du et al., 2024; Chen et al., 2024; Smit et al., 2024). A further extension of multi-agent debates is to model human deliberation in domains where no objective ground truth exists, including politics, ethics, peer review, and stakeholder decision-making (Park et al., 2024a; 2023; Anthis et al., 2025; Aher et al., 2023; Taubenfeld et al., 2024). In these settings, the goal is to study phenomena such as how beliefs evolve, influence accumulates, and groups reason under uncertainty. In domains such as city planning and politics simulations, these debates are often used to explore policy trade-offs or anticipate stakeholder reactions before real-world deliberation, making the reliability of their outputs consequential (Park et al., 2023; Argyle et al., 2023).

Despite this increased interest in multi-agent LLMs for real-world debates, evaluation practices for the effectiveness of LLM debates have largely remained outcome-focused. Most existing work relies on proxy measures such as debate length, LLM-as-judge scores, human judgments of persuasiveness or realism, and whether consensus/majority vote is reached (Park et al., 2023; Chen et al., 2024; Zhang et al., 2024; Sternlicht et al., 2025). These proxies are appealing because they are easy to compute and intuitive to interpret, but they implicitly assume that desirable outcomes such as agreement or LLM scores are sufficient indicators of correctness or reliability.

However, recent studies in LLMs showed that the outcome-based metrics are not sufficient for real-world debate evaluations. Prior work has shown that consensus and majority agreement can arise from sycophancy, imitation, or dominance rather than genuine deliberation (Janis, 1972; Sunstein, 2005; Pitre et al., 2025; Taubenfeld et al., 2024). Similarly, LLM-as-judge evaluations are sensitive to prompt framing, model biases, and surface-level cues, and often collapse rich deliberative structure into a single subjective score (Sternlicht et al., 2025; Chen et al., 2024). Human annotation is expensive, with high variance and potentially biased depending on the task. Together, these findings show that commonly used outcome-based proxies are not efficient or automatic, and cannot be reliably trusted to reflect the correctness of the outcome or deliberation quality. This

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

limitation is concerning in both objective and subjective dataset settings. In objective settings where ground truth exists, performance degradation is often addressed by optimizing for agreement—such as enforcing consensus or majority voting (Chen et al., 2024; Pitre et al., 2025) without resolving the underlying interaction failures that caused the error. If consensus arises out of sycophancy or agent domination, optimizing for consensus further worsens the issue. In subjective or open-ended domains without ground truth, this reliance on outcome-based proxies is especially problematic. If the deliberative process itself is flawed, then any outcome it produces is inherently unreliable, regardless of whether it appears coherent or convincing.

This paper is motivated by a simple hypothesis: a debate should be evaluated with both, the outcome of the debate and the process that produced it. Prior work on deliberative democracy explicitly argues that debate quality should be evaluated based on procedural properties - such as reciprocity, justification, and responsiveness rather than on agreement or other outcome-based signals (Habermas, 1991; Gutmann & Thompson, 1996; Fishkin, 2009; Mansbridge et al., 2012). In this work, we introduce a diagnostic evaluation framework for multi-agent debates that explicitly separates deliberative process from debate outcomes. We formalize a minimal set of axioms for debate quality grounded in deliberative theory and operationalize them as interpretable, process-level metrics that capture engagement, responsiveness, influence asymmetry, and balance. These diagnostics are designed to detect concrete interaction failures, such as dogmatism, non-reciprocal updating, domination, and premature convergence, without requiring access to ground truth. We evaluate the proposed framework across both objective benchmarks with known answers and real-world domains without ground truth, comparing it against commonly used outcome-based proxies. Our experiments show that our metrics (1) align better with both correctness and human preferences, (2) helps us understand consensus better, and (3) optimizing towards our metrics using agent prompts yields higher accuracy.

2. Related Work

Multi-Agent LLM Debate Simulations In objective datasets with ground truth, when multi-agent debate systems fail to improve accuracy, analysis and optimization often focus on outcome-level signals such as increasing agreement or enforcing consensus rather than diagnosing failures in the underlying interaction dynamics. This is done through several methods like prompt optimization (Pitre et al., 2025) or confidence weighing (Chen et al., 2024). As a result, debugging efforts tend to optimize for convergence behavior rather than addressing whether agents are meaningfully engaging, responding, or updating their beliefs, despite interaction being the core mechanism these systems are intended to improve.

In subjective domains without ground truth, such as politics, ethics, or city planning, where accuracy-based evaluation is unavailable, this challenge becomes even more prominent. In these settings, existing work uses consensus, majority vote, or other proxies as evaluation (Park et al., 2024b; Chen et al., 2025; Zhang et al., 2024; Chen et al., 2024; Taubenfeld et al., 2024). Current mechanisms cannot reveal if the underlying cause of consensus is healthy deliberation or interaction based issues like sycophancy or dominance (Taubenfeld et al., 2024; Oh et al., 2025; Pitre et al., 2025; tse Huang et al., 2025).

In contrast to prior work, we focus on evaluating debate quality through explicit, goal-grounded behavioral metrics that directly measure interaction dynamics, rather than treating outcomes such as consensus or expensive human annotations as proxies for deliberation quality.

Diagnostic Metrics in the Absence of Ground Truth In domains with no ground truth, prior work has adopted evaluation frameworks grounded in first principles rather than outcome optimization. In economics and social choice, collective mechanisms are evaluated by specifying axioms—such as Pareto efficiency, monotonicity, and anonymity, and analyzing which properties they satisfy or violate, often using constructed examples or simulations to reveal failure modes (Arrow et al., 1964; Sen, 1970). These frameworks emphasize diagnostic criteria that explain why a process succeeds or fails, rather than collapsing behavior into a single scalar objective.

A similar diagnostic approach has been adopted in machine learning settings where ground truth explanations do not exist. Work on explainability proposes axiomatic or property-based criteria—such as faithfulness, stability, and consistency—and evaluates them by probing models under controlled perturbations, demonstrating their ability to detect known failure modes rather than match a gold standard (Sundararajan et al., 2017; Yoshikawa et al., 2024; Atanasova et al., 2020).

Our work follows this tradition by introducing a small, goal-grounded set of behavioral diagnostics for multi-agent debate. These metrics are derived from explicit deliberative principles, validated through targeted pathologies, and shown to provide insight into debate dynamics beyond what outcome-based measures can capture.

3. Diagnostic Properties

We now define a set of diagnostic properties of a multi-agent debate, and propose how to quantify them.

3.1. Axioms for Debate Quality

In settings without ground truth, debate quality cannot be evaluated using correctness alone. Prior work in deliberative democracy, social choice, and collective decision-making argues that agreement or convergence is insufficient to characterize the quality of deliberation. Instead, these traditions evaluate deliberation in terms of procedural properties that describe how participants exchange reasons, respond to one another, and exert influence during discussion (Habermas, 1991; Gutmann & Thompson, 1996; Mansbridge et al., 2012). While this literature does not define a single notion of a "good" debate, it exhibits broad agreement on a recurring set of normative requirements. We draw on these ideas to define a minimal set of axioms that are both theoretically grounded and operationalizable in multi-agent systems.

Monotonicity: A core assumption across deliberative accounts is that deliberation is a process of belief updating: participants are expected to revise their positions in response to arguments presented during interaction, rather than merely restating fixed views (Habermas, 1991; List & Pettit, 2011). This motivates monotonicity, which requires that agents exhibit meaningful belief change over the course of a debate.

Responsiveness: This axioms argues the expectation that belief updates should be causally linked to others' arguments rather than arising independently or coincidentally (Gutmann & Thompson, 1996; Mansbridge et al., 2012).

Non-domination: A set of requirements concerns the distribution of influence. Deliberative quality degrades when a single participant disproportionately controls the discussion or outcome (Pettit, 1997; Sunstein, 2005). This motivates non-domination, which requires influence to be distributed rather than concentrated.

Pluralism: Deliberation is valued for its epistemic benefits precisely because it preserves diversity of perspectives. Premature convergence or imitation can undermine these benefits by collapsing independent reasoning processes (Sunstein, 2005; Page, 2025; Landemore, 2012). This motivates pluralism, which requires the maintenance of meaningful disagreement during deliberation rather than rapid or superficial consensus. In this work, we consider scenarios where there is no social hierarchy in agents, and each agent should get an equal say in the debate. If there are hierarchies in the agent, they can be further modeled in future work.

Openness: In a debate, participants are expected to engage with counterarguments rather than exhibiting dogmatism or strategic entrenchment (Dewey, 2021; Habermas, 1991; List & Pettit, 2011)

Conditional Convergence: Agreement—when it occurs—should emerge through reasoned interaction rather than coercion, fatigue, or conformity. This concern motivates conditional convergence, which treats consensus as desirable only when it arises from sustained, healthy deliberation (Janis, 1972; Mansbridge et al., 2012).

Stability: Debate depends on both the process and the output, and hence, we include outcome-level axioms that ensure agent utility. Stability requires that outcomes be robust to unilateral deviation, corresponding to incentive compatibility or Nash equilibrium (Nash, 1950).

Agent Utility: Group welfare captures the requirement that collective decisions not systematically disadvantage subsets of participants, even when agreement is achieved (Arrow et al., 1964; Sen, 1970).

These axioms are not intended to provide a complete theory of deliberation. Rather, they define a minimal set of constraints that recur across multiple ground-breaking communication, debate and economics works that define ideal interaction, and can be linked to observable interaction-level failures in multi-agent debates. In the following section, we show how violations of these axioms manifest as concrete behavioral pathologies, which we operationalize as diagnostic metrics.

3.2. From Axioms to Diagnostic Metrics

Axioms are ideal qualities in debates. To operationalize them, we adopt a diagnostic perspective: for each axiom, we identify concrete interaction-level failure modes that arise when the axiom is violated, and then define metrics that detect these failures using simple, interpretable equations computed from debate interactions.

Notebly, in our case, the mapping from axioms to metrics is many-to-one because multiple axioms may give rise to similar observable failures. For example, both openness and monotonicity can be violated when agents repeatedly ignore others' contributions—and can therefore captured by a single diagnostic metric. To acquire a minimum nonredundant set of constraints, we defined four metrics, explained in detail below. This many-to-one mapping of deriving metrics from axioms is a common practice in both computational and social science domains (Sana et al., 2025; Revel et al., 2025). Each metric detects a specific interaction error rather than producing a global quality score. Taken together, these diagnostics provide complementary views of debate quality by revealing how a debate fails, not merely whether it converges.

3.3. Process-Level Diagnostic Metrics

We now introduce a set of process-level diagnostic metrics that translate the deliberative axioms from Section 3.1 into observable properties of debate dynamics. Each metric is motivated by a characteristic failure mode that arises when one or more axioms are violated, and is computed directly from debate transcripts. Importantly, these diagnostics rely only on agents' stance trajectories and the quality of their stated reasoning; they do not require ground truth labels or access to the final outcome.

Notation A debate consists of A agents interacting over T rounds. At each round t, agent a expresses a stance

$$x_a^t \in \{-2, -1, 0, 1, 2\}, \tag{1}$$

corresponding to strongly disagree through strongly agree.

Each agent message is associated with a reasoning quality score as derived from (Prasad et al., 2023)

$$q_a^t \in [0, 1]. \tag{2}$$

Let

$$\mu^t = \frac{1}{A} \sum_{a=1}^{A} x_a^t \tag{3}$$

denote the mean stance at round t, and let μ_{-a}^t denote the mean stance of all agents except a.

Engagement The axioms of monotonicity and openness characterize deliberation as a learning process: agents should remain receptive to opposing arguments and revise their beliefs when presented with compelling reasons. These axioms are violated when agents either remain dogmatic, repeatedly asserting the same stance, or exhibit arbitrary belief changes without substantive justification. In both cases, the debate fails to exhibit sustained, reasoned belief updating over time.

To measure this, we require a signal that captures both whether agents revise their stances across rounds and whether those revisions are supported by meaningful reasoning. We therefore define engagement as the average magnitude of stance change, weighted by the quality of the accompanying justification:

$$\text{Engagement} = \frac{1}{A(T-1)} \sum_{a=1}^{A} \sum_{t=2}^{T} q_a^t \cdot |x_a^t - x_a^{t-1}|. \tag{4}$$

Low engagement indicates stagnation or ungrounded volatility, reflecting violations of monotonicity or openness, while higher values indicate that agents remain receptive and update their beliefs in a reasoned manner throughout the debate.

Responsiveness The axioms of monotonicity and responsiveness require belief updates to arise through interaction: agents should revise their stances *in response to* others' arguments rather than through parallel or independent reasoning. These axioms are violated when agents update their beliefs without meaningfully engaging with one another, producing non-reciprocal or weakly coupled belief change. In such cases, stance movement may occur, but it fails to reflect deliberative exchange.

To measure this, we require a signal that captures whether an agent's belief updates move in the direction of the group in response to prior discussion, rather than independently. We approximate reciprocal influence by checking whether an agent's stance moves closer to the rest of the group relative to the previous round, weighting updates by reasoning quality to exclude unsupported shifts. Formally, we define the indicator:

$$\Delta_a^t = \nVdash\left(|x_a^t - \mu_{-a}^{t-1}| < |x_a^{t-1} - \mu_{-a}^{t-1}|\right).$$

Low responsiveness indicates belief updates that are weakly coupled to deliberative exchange, while higher values indicate reciprocal, interaction-driven belief revision throughout the debate.

Influence Asymmetry (Non-Domination) The axiom of non-domination requires that deliberation not devolve into an authoritarian dynamic in which a single agent disproportionately shapes the beliefs of others. While such domination may produce rapid convergence or apparent consensus, it reflects a failure of deliberative equality rather than successful collective reasoning. Identifying this failure mode therefore requires measuring how influence is distributed across participants.

To do so, we estimate an agent's influence by counting how often other agents move closer to that agent's prior stance, conditional on justified belief updates. The influence exerted by agent a is defined as:

$$I_a = \sum_{b \neq a} \sum_{t=2}^{T} q_b^t \cdot \nVdash\left[|x_b^t - x_a^{t-1}| < |x_b^{t-1} - x_a^{t-1}|\right]. \tag{5}$$

Normalizing these values yields a distribution over agents,

$$p_a = \frac{I_a}{\sum_{k=1}^{A} I_k}, \tag{6}$$

from which we compute influence asymmetry as the normalized concentration of influence:

$$\text{Asymmetry} = 1 - \frac{H(\mathbf{p})}{\log A}, \quad H(\mathbf{p}) = -\sum_{a=1}^{A} p_a \log p_a. \tag{7}$$

High asymmetry indicates violations of non-domination, with influence concentrated in a small subset of agents, while low asymmetry reflects more evenly distributed deliberative influence.

Balance The axioms of pluralism and conditional convergence constrain the global trajectory of deliberation. Pluralism requires that meaningful disagreement be preserved, while conditional convergence permits agreement only when it emerges gradually through deliberation. Both axioms are violated by degenerate dynamics, including instantaneous collapse to consensus or highly unstable oscillations in disagreement.

To characterize these behaviors, we examine the temporal evolution of stance dispersion. Let σ^t denote the standard deviation of stances at round t. Total convergence over the debate is defined as

$$C_{\text{total}} = \sigma^1 - \sigma^T, \tag{8}$$

and the largest single-round convergence as

$$C_{\max} = \max_{t \geq 2} \left(\sigma^{t-1} - \sigma^t \right). \tag{9}$$

Premature collapse corresponds to a large ratio $C_{\max}/C_{\text{total}}$. To capture instability, we measure volatility as the number of reversals in the direction of dispersion change:

$$V = \sum_{t=3}^{T} \mathbb{1}\Big[(\sigma^{t-2} - \sigma^{t-1}) \cdot (\sigma^{t-1} - \sigma^t) < 0\Big]. \tag{10}$$

We define balance as

$$\text{Balance} = \left(1 - \frac{C_{\max}}{C_{\text{total}} + \varepsilon}\right) \cdot \left(1 - \frac{V}{T-2}\right), \tag{11}$$

where ε avoids division by zero. High balance reflects deliberative trajectories that preserve diversity while allowing gradual convergence, whereas low balance indicates collapse or instability inconsistent with conditional convergence.

3.4. Outcome-Level Diagnostic Metrics

In addition to process-level diagnostics, we consider two outcome-level properties that are commonly studied in economic and game-theoretic analyses of collective processes. These diagnostics do not assess how deliberation unfolds, but rather evaluate properties of the terminal state produced by the debate. Importantly, they are treated as necessary but not sufficient conditions for deliberative quality: a debate may satisfy these outcome constraints while still exhibiting severe process-level failures.

Stability The axiom of stability requires that a deliberative outcome be robust to unilateral deviation. An outcome is unstable if at least one agent would prefer to change its final stance given the stances of others, indicating that agreement—if present—does not reflect genuine conviction. This instability is a result of agents colluding strategically or converging due to pressure rather than persuasion.

We operationalize stability by testing whether the final stance profile constitutes a Nash equilibrium under agent utilities. Let $u_a(x_a, x_{-a})$ denote the utility of agent a given its own stance and the stances of others. An outcome is considered stable if no agent can increase its utility by unilaterally deviating from its final stance:

$$u_a(x_a^T, x_{-a}^T) \geq u_a(x'_a, x_{-a}^T) \quad \forall a,\ \forall x'_a \in \mathcal{X}. \tag{12}$$

We define the stability score as the fraction of agents for which this condition holds. High stability indicates outcomes that are internally consistent with agent preferences, while low stability reveals outcomes sustained only through transient or strategic compliance.

Group Welfare The axiom of group welfare requires that deliberation not systematically disadvantage subsets of agents. This axiom is violated when agents are unhappy with the outcomes of the debate.

We operationalize group welfare by comparing agents' utilities before and after deliberation. Let $u_a^1 = u_a(x_a^1, x_{-a}^1)$ and $u_a^T = u_a(x_a^T, x_{-a}^T)$ denote agent a's utility at the beginning and end of the debate, respectively. Group welfare is measured as the average change in utility:

$$\text{Welfare} = \frac{1}{A} \sum_{a=1}^{A} \left(u_a^T - u_a^1 \right). \tag{13}$$

In addition to this aggregate score, we analyze the distribution of utility changes across agents to identify uneven or asymmetric welfare effects. Welfare is not treated as a correctness signal, but as a diagnostic lens on the distributional consequences of deliberation outcomes. Prompting is used to ask the agent about their happiness/satisfaction with the given statement.

4. Experiments

4.1. Datasets

We demonstrate our agent creation and multi-agent debate methods using both an objective benchmark and a real-world domains without ground truth.

For an objective dataset, we use MMLU dataset (Hendrycks et al., 2020). For subjective datasets, we use city planning stakeholder interviews (Gosain et al., 2022) and politics simulation interviews from (C-SPAN, 2025). To initiate agents, we use persona assignment based on these interviews. For each interviewee, we instruction-tune the model to act as that individual, using the interview questions and responses to condition the agent's behavior. For debate experiments, we generate a large set of policy-style questions on which agents express their stances using five-point Likert ratings (1–5). Details about this can be found in Appendix A.2.

4.2. Models

We run all experiments using three large language models to assess generalization across model families: GPT-4 (gpt-

4o), Gemini 1.5 Pro, and Qwen2.5-72B-Instruct. For each dataset and experimental setting, we use the same agent construction, prompting, and debate protocols across models, with no model-specific tuning beyond standard inference parameters.

4.3. Experimental Setup

All experiments follow a fixed multi-agent debate protocol with A agents and $T = 5$ rounds. In round 0, agents independently produce an initial answer and explanation. In subsequent rounds $t = 1, \ldots, 5$, agents are prompted to interact with other agents and update their stance as needed, following the interaction protocol of Liang et al. (2024).

After the final round, if there is no consensus, a moderator is shown the full debate history and asked to select a final answer. This moderator-selected answer is used to compute an answer, while our diagnostic metrics are applied separately to evaluate the quality of the deliberative process that produced it. Additional implementation details can be found in Appendix A.3.

Correlation with accuracy and human preference We evaluate the relationship between interaction process signals and outcome quality in both objective and subjective settings. For the MMLU dataset, we compute the Spearman rank correlation between baselines (LLM-as-judge, number of rounds, number of tokens), our debate-level diagnostic scores and final-answer accuracy.

For subjective tasks, we evaluate diagnostics using pairwise agreement with human preference judgments. Three communication major annotators are shown pairs of full debate transcripts and asked to indicate which debate exhibits higher-quality deliberation, without access to agent identities or experimental conditions. For each diagnostic and baseline, we report pairwise accuracy: the fraction of debate pairs for which the diagnostic correctly predicts the debate preferred by human annotators (chance = 0.5). Inter-annotator agreement is reported in Section 5.

Sanity checks via controlled violations To verify that each diagnostic responds selectively to the deliberative property it is designed to capture, we conduct controlled experiments in which specific axioms are intentionally violated through targeted prompting. Agents are prompted to exhibit behaviors such as dogmatism, non-responsiveness, domination, sycophantic belief change, oscillation, or premature convergence, while all other aspects of the debate protocol are held fixed. This is verified by human annotators. We then measure how each diagnostic degrades under its corresponding violation, checking for specificity rather than global sensitivity.

Causal perturbation experiments Finally, we conduct causal perturbation experiments to test whether changes in diagnostic scores correspond to meaningful changes in debate outcomes. Individual agents are prompted to systematically increase or decrease behaviors associated with specific metrics (e.g., increased dogmatism, heightened responsiveness). We verify that the targeted diagnostic shifts in the intended direction, bin debates by resulting score ranges, and analyze how final-answer accuracy varies across these bins. This allows us to study how deliberative process properties relate to downstream performance under controlled interventions.

5. Results

Process-Level Signals Predict Correctness on Objective Tasks We evaluate whether debate evaluation methods align with correctness on objective tasks with ground truth. As shown in Table 1, commonly used outcome-based proxy (LLM-as-judge) exhibits only moderate correlation with accuracy, while debate length and number of tokens are particularly weak signals.

Outcome-level signals such as stability and group welfare have higher, but still a moderate correlation with accuracy. In contrast, all proposed process-level diagnostics show substantially stronger alignment with correctness across model families. This shows that our metrics are correlated with accuracy- for example, when engagement between agents increases, so does accuracy. This insight is crucial because it shows that correctness is not primarily a function of debate outcomes or surface-level signals, but of the underlying interactions between the agents. It also shows why outcome-based evaluations are not sufficient: they cannot capture nuances about the interaction like process based metrics do.

Process-Level Signals Align with Human Preferences on Subjective Tasks Human preference serves as a soft ground truth in the absence of ground truth for subjective datasets. We achieve a Cohen's Kappa score of 0.73, which is considered high for this task (Gretz et al., 2020) Table 2 shows that humans do not prefer debates that are longer, more verbose, or scored highly by an LLM judge—these signals perform only slightly above chance. Outcome-level measures such as stability and agent utility do somewhat better, but still fall short of capturing what humans view as high-quality deliberation.

In contrast, process-level diagnostics align strongly with human preferences across all model families. Metrics that capture how agents interact—especially whether influence is evenly distributed, whether agents respond to one another, and whether disagreement resolves gradually—consistently perform best. Influence asymmetry stands out as the strongest signal, suggesting that humans are particularly

Table 1. Spearman correlation (ρ) between debate-level signals and answer correctness on objective benchmarks across three model families.

Signal / Metric	GPT-4o	Qwen2.5	Gemini-1.5
Num. tokens	0.21	0.18	0.20
Num. Rounds	0.24	0.22	0.23
LLM-as-Judge	0.59	0.55	0.57
Nash-Stability	0.60	0.58	0.59
Agent Utility	0.66	0.62	0.64
Engagement	0.69	0.69	0.71
Responsiveness	0.71	0.67	0.70
Balance	0.71	0.68	0.70
Influence Asym. (inv.)	0.75	0.71	0.72
Avg. Process Metrics	**0.76**	**0.73**	**0.75**

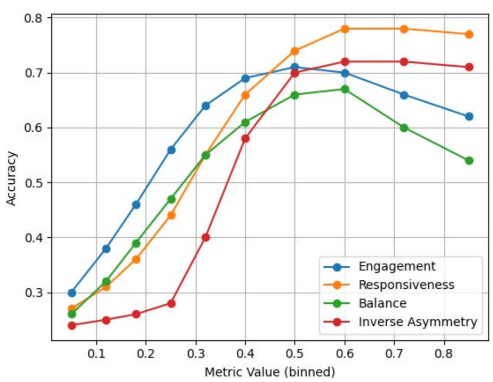


Figure 1. Causal Perturbations link process quality to outcomes

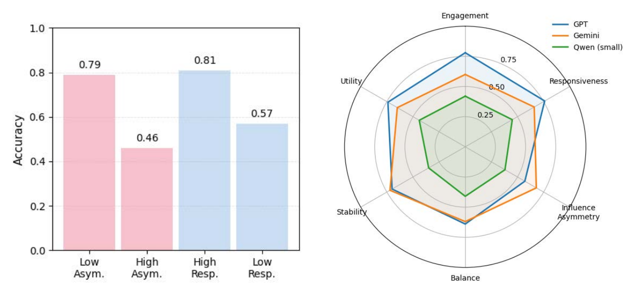
sensitive to domination and coercive dynamics. Overall, these results indicate that human judgments of debate quality are driven by interaction structure rather than surface fluency or final agreement. This shows the power of our metrics, which capture human preference without having to go through an expensive, difficult human annotation procedure.

Sanity Tests Validate Metric Specificity Table 3 shows that each diagnostic degrades most strongly when the deliberative property it is designed to capture is explicitly violated. Engagement drops sharply under dogmatism, responsiveness degrades when agents ignore one another, influence asymmetry increases under domination, and balance degrades under oscillation or premature convergence.

Importantly, metrics do not degrade uniformly across all interventions. For example, domination primarily affects influence asymmetry, while premature convergence selectively impacts balance. This specificity indicates that the diagnostics are not generic proxies for debate noise or difficulty, but capture distinct and interpretable failure modes of deliberation. Sycophancy is captured primarily by engagement, but other metrics also capture a part of this issue. Together, these results support the construct validity of the proposed metrics.

Table 2. Pairwise accuracy with human preferences on subjective tasks across three model families. Values indicate the fraction of debate pairs for which the metric correctly predicts the debate preferred by human annotators (chance = 0.5).

Signal / Metric	GPT-4o	Qwen2.5	Gemini-1.5
Num. tokens	0.51	0.50	0.51
Rounds	0.51	0.50	0.51
LLM-as-Judge	0.62	0.59	0.61
Nash-stability	0.64	0.61	0.63
Agent Utility	0.66	0.63	0.65
Engagement	0.68	0.65	0.67
Responsiveness	0.74	0.71	0.73
Balance	0.73	0.70	0.72
Influence Asymm. (inv.)	0.89	0.86	0.88
All Process Metrics	**0.78**	**0.75**	**0.77**



(a) Consensus vs. process quality

(b) Effect of model size

Figure 2. a. Demonstrating why consensus is a weak metric which is unable to capture nuances; b. Different models and their metric scores

Causal Perturbations Link Process Quality to Outcomes We conduct causal perturbation experiments by explicitly prompting agents to increase or decrease behaviors associated with each process-level diagnostic, and bin debates by the resulting metric values. Figure 1 shows that all diagnostics exhibit a similar pattern: accuracy improves as the corresponding process property increases from low to moderate levels, but degrades when pushed too far.

Engagement improves accuracy as agents revise their beliefs in response to evidence, peaking at moderate values. Beyond this point, excessive belief revision reduces accuracy, suggesting over-updating or loss of stable commitments. Responsiveness shows a strong positive effect up to its peak, after which accuracy plateaus and slightly declines, indicating that constant reaction without independent reasoning can become counterproductive.

Balance exhibits a clear optimum at intermediate values. Debates with very low balance collapse prematurely, while highly volatile debates fail to converge, both leading to lower accuracy. Influence asymmetry (inverse) improves accuracy by preventing domination, but extreme egalitarianism eventually degrades performance, likely by suppressing

Table 3. Sanity (construct validity) tests on OpenAI models. Values report the magnitude of metric degradation ($\Delta = M_{\text{unbroken}} - M_{\text{broken}}$) under targeted interventions that violate specific deliberation properties. Each metric degrades most strongly when its corresponding property is broken.

Broken Property (Intervention)	Engagement	Responsiveness	Asymmetry (inv.)	Balance
Dogmatism / No Engagement (agents do not change positions)	**0.81**	0.18	0.05	0.22
No Responsiveness (agents ignore each other)	0.21	**0.79**	0.07	0.19
Sycophantic Change (agreement without evidence)	0.64	0.31	0.42	0.38
Dictator Agent (one agent dominates)	0.26	0.34	**0.83**	0.29
Oscillating Views (inconsistent position flipping)	0.58	0.52	0.11	**0.55**
Premature Convergence (early collapse to one view)	0.33	0.27	0.24	**0.76**

strong arguments and slowing convergence.

These results show that effective deliberation lies in a constrained regime: learning, interaction, balance, and influence must be present, but not maximized. Process-level diagnostics therefore capture causal mechanisms that explain when and why debates succeed, rather than serving as monotonic optimization targets.

6. Discussion

Why is Consensus a Weak Measure? Figure 2(a) demonstrates that consensus alone is a poor indicator of correctness. We select debates that reached consensus and stratify them by process quality, separating the top and bottom quartiles according to specific diagnostics. Despite identical outcomes (consensus reached), accuracy differs sharply across these groups. For example, debates with low influence asymmetry achieve substantially higher accuracy than those with high asymmetry, and debates with high responsiveness are far more accurate than those with low responsiveness. This shows that agreement can be reached through fundamentally different interaction dynamics—some that support correct reasoning and others that do not. Consensus-based evaluation collapses these cases, while process-level diagnostics distinguish informative agreement from convergence driven by domination or weak interaction.

Effect of Model Size Figure 2(b) shows how deliberative process properties vary across model sizes. Larger models consistently exhibit higher engagement and responsiveness, reflecting greater capacity to revise beliefs and respond to other agents' arguments. However, gains are not uniform across all diagnostics. Influence asymmetry remains substantial even for larger models, indicating that increased capability does not prevent domination dynamics. Balance also improves only modestly, suggesting that larger models may still converge prematurely or fail to sustain productive disagreement. Smaller models exhibit uniformly lower scores across most metrics, but do not necessarily perform worse on influence balance. Overall, these results show that scaling model size improves some aspects of deliberation but does not guarantee high-quality interaction, reinforcing the need for process-level evaluation beyond model size or aggregate performance.

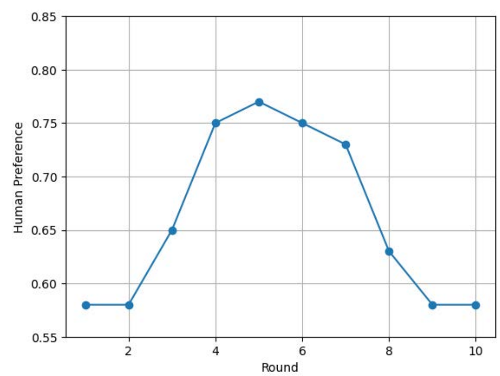


Figure 3. Number of rounds vs human-preference

6.1. Number of Rounds and Debate Quality

In Figure 3, we plot the number of debate rounds against accuracy on human preference scores in the city planning setting. The resulting curve exhibits a consistent U-shaped pattern. Humans prefer medium sized debates, as very short debates tend to underperform and very long ones show repeating, or unsubstantial conversations. This pattern implies that productive deliberation occurs within a bounded window of interaction, after which additional rounds yield diminishing returns.

7. Conclusion

This work shows that evaluating multi-agent debates requires looking beyond agreement and surface-level outcomes. Process-level diagnostics reveal how debates succeed or fail by exposing underlying interaction dynamics that outcome-based measures cannot capture. Together, these results motivate a shift toward diagnostic, process-aware evaluation of deliberative LLM systems.

8. Impact Statement

Multi-agent LLM debates are increasingly used to inform decisions in high-stakes domains such as policy analysis, urban planning, and political simulation, where errors or misleading conclusions can have real-world consequences. This work contributes a diagnostic framework that helps distinguish meaningful deliberation from superficial agreement, reducing the risk of relying on debates that appear reliable but are driven by dominance or sycophancy. By shifting evaluation from outcome-only measures to interpretable process-level diagnostics, our approach supports more transparent, accountable, and robust use of deliberative LLM systems. While the framework does not eliminate bias or guarantee correctness, it provides practical tools for detecting interaction failures before they propagate into downstream decisions. We hope this work encourages more careful evaluation practices as multi-agent LLM systems are deployed in real-world decision-making settings.

References

Aher, G., Arriaga, R. I., and Kalai, A. T. Using large language models to simulate multiple humans and replicate human subject studies. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023.

Anthis, J. R., Liu, R., Richardson, S. M., Kozlowski, A. C., Koch, B., Brynjolfsson, E., Evans, J., and Bernstein, M. S. Position: Llm social simulations are a promising research method. In *Forty-second International Conference on Machine Learning Position Paper Track*, 2025.

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31 (3):337–351, February 2023. ISSN 1476-4989. doi: 10.1017/pan.2023.2. URL http://dx.doi.org/10.1017/pan.2023.2.

Arrow, K. J. et al. *Social choice and individual values*, volume 2. Wiley New York, 1964.

Atanasova, P., Simonsen, J. G., Lioma, C., and Augenstein, I. A diagnostic study of explainability techniques for text classification. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pp. 3256–3274, 2020.

C-SPAN. C-span video library. https://www.c-span.org/, 2025. Accessed: 2025-10-05.

Chen, H., Ji, W., Xu, L., and Zhao, S. Multi-agent consensus seeking via large language models, 2025. URL https://arxiv.org/abs/2310.20151.

Chen, J., Saha, S., and Bansal, M. ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs. In Ku, L.-W., Martins, A., and Srikumar, V. (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7066–7085, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.381. URL https://aclanthology.org/2024.acl-long.381/.

Dewey, J. 9. creative democracy—the task before us. In *America's Public Philosopher*, pp. 59–66. Columbia University Press, 2021.

Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., and Mordatch, I. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024.

Fishkin, J. *When the people speak: Deliberative democracy and public consultation*. Oup Oxford, 2009.

Gosain, P., Zhang, L., and Ganapati, N. E. Understanding multisector stakeholder value systems on housing resilience in the city of miami. *International Journal of Disaster Risk Reduction*, 77:103061, 2022.

Gretz, S., Friedman, R., Cohen-Karlik, E., Toledo, A., Lahav, D., Aharonov, R., and Slonim, N. A large-scale dataset for argument quality ranking: Construction and analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 7805–7813, 2020.

Gutmann, A. and Thompson, D. *Democracy and Disagreement*. Harvard University Press, 1996.

Habermas, J. *The Theory of Communicative Action, Vol. 1: Reason and the Rationalization of Society*. Beacon Press, 1991.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Innes, J. E. and Booher, D. E. Consensus building and complex adaptive systems. *Journal of the American Planning Association*, 65(4):412–423, 1999. doi: 10.1080/01944369908976071. URL https://doi.org/10.1080/01944369908976071.

Janis, I. L. *Victims of Groupthink: A Psychological Study of Foreign-Policy Decisions and Fiascoes*. Houghton Mifflin, 1972.

Landemore, H. Democratic reason: Politics, collective intelligence, and the rule of the many. 2012.

Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Shi, S., and Tu, Z. Encouraging divergent thinking in large language models through multi-agent debate. In Al-Onaizan, Y., Bansal, M., and Chen, Y.-N. (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 17889–17904, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.992. URL https://aclanthology.org/2024.emnlp-main.992/.

List, C. and Pettit, P. *Group agency: The possibility, design, and status of corporate agents*. Oxford University Press, 2011.

Mansbridge, J., Bohman, J., Chambers, S., Christiano, T., Fung, A., Parkinson, J., Thompson, D. F., and Warren, M. E. A systemic approach to deliberative democracy. *Deliberative systems: Deliberative democracy at the large scale*, pp. 1–26, 2012.

Nash, J. F. Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences of the United States of America*, 36(1):48–49, 1950.

Oh, J., Jeong, M., Ko, J., and Yun, S.-Y. Understanding bias reinforcement in llm agents debate, 2025. URL https://arxiv.org/abs/2503.16814.

Page, S. The difference: How the power of diversity creates better groups, firms, schools, and societies. 2025.

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, UIST '23, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701320. doi: 10.1145/3586183.3606763. URL https://doi.org/10.1145/3586183.3606763.

Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Willer, R., Liang, P., and Bernstein, M. S. Generative agent simulations of 1,000 people, 2024a. URL https://arxiv.org/abs/2411.10109.

Park, S., Kim, J., Jin, S., Park, S., and Han, K. PREDICT: Multi-agent-based debate simulation for generalized hate speech detection. In Al-Onaizan, Y., Bansal, M., and Chen, Y.-N. (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 20963–20987, Miami, Florida, USA, November 2024b. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.1166. URL https://aclanthology.org/2024.emnlp-main.1166/.

Pettit, P. Republicanism: a theory of freedom and government. *Clarendon P*, 1997.

Pitre, P., Ramakrishnan, N., and Wang, X. CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation. In Che, W., Nabende, J., Shutova, E., and Pilehvar, M. T. (eds.), *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 22112–22133, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.1141. URL https://aclanthology.org/2025.findings-acl.1141/.

Prasad, A., Saha, S., Zhou, X., and Bansal, M. Receval: Evaluating reasoning chains via correctness and informativeness. *arXiv preprint arXiv:2304.10703*, 2023.

Revel, M., Milli, S., Lu, T., Watson-Daniels, J., and Nickel, M. Representative ranking for deliberation in the public sphere. *arXiv preprint arXiv:2503.18962*, 2025.

Sana, S., Brous, D., Wells, M. T., and Duchin, M. Quantitative relaxations of arrow's axioms, 2025. URL https://arxiv.org/abs/2506.12961.

Sen, A. *Collective Choice and Social Welfare*. Harvard University Press, Cambridge, MA, 1970.

Smit, A., Grinsztajn, N., Duckworth, P., Barrett, T. D., and Pretorius, A. Should we be going mad? a look at multi-agent debate strategies for llms. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024.

Sternlicht, N., Gera, A., Bar-Haim, R., Hope, T., and Slonim, N. Debatable intelligence: Benchmarking llm judges via debate speech evaluation, 2025. URL https://arxiv.org/abs/2506.05062.

Sundararajan, M., Taly, A., and Yan, Q. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning - Volume 70*, ICML'17, pp. 3319–3328. JMLR.org, 2017.

Sunstein, C. R. *Why Societies Need Dissent*. Harvard University Press, Cambridge, MA, 2005.

Taubenfeld, A., Dover, Y., Reichart, R., and Goldstein, A. Systematic biases in LLM simulations of debates. In Al-Onaizan, Y., Bansal, M., and Chen, Y.-N. (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 251–267, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.16. URL https://aclanthology.org/2024.emnlp-main.16/.

tse Huang, J., Zhou, J., Jin, T., Zhou, X., Chen, Z., Wang, W., Yuan, Y., Lyu, M. R., and Sap, M. On the resilience of llm-based multi-agent collaboration with faulty agents, 2025. URL https://arxiv.org/abs/2408.00989.

Yoshikawa, Y., Kimura, M., Shimizu, R., and Saito, Y. Explaining black-box model predictions via two-level nested feature attributions with consistency property. *arXiv preprint arXiv:2405.14522*, 2024.

Zhang, Y., Yang, X., Feng, S., Wang, D., Zhang, Y., and Song, K. Can llms beat humans in debating? a dynamic multi-agent framework for competitive debate, 2024. URL https://arxiv.org/abs/2408.04472.

A. Appendix

A.1. Intuition Figure

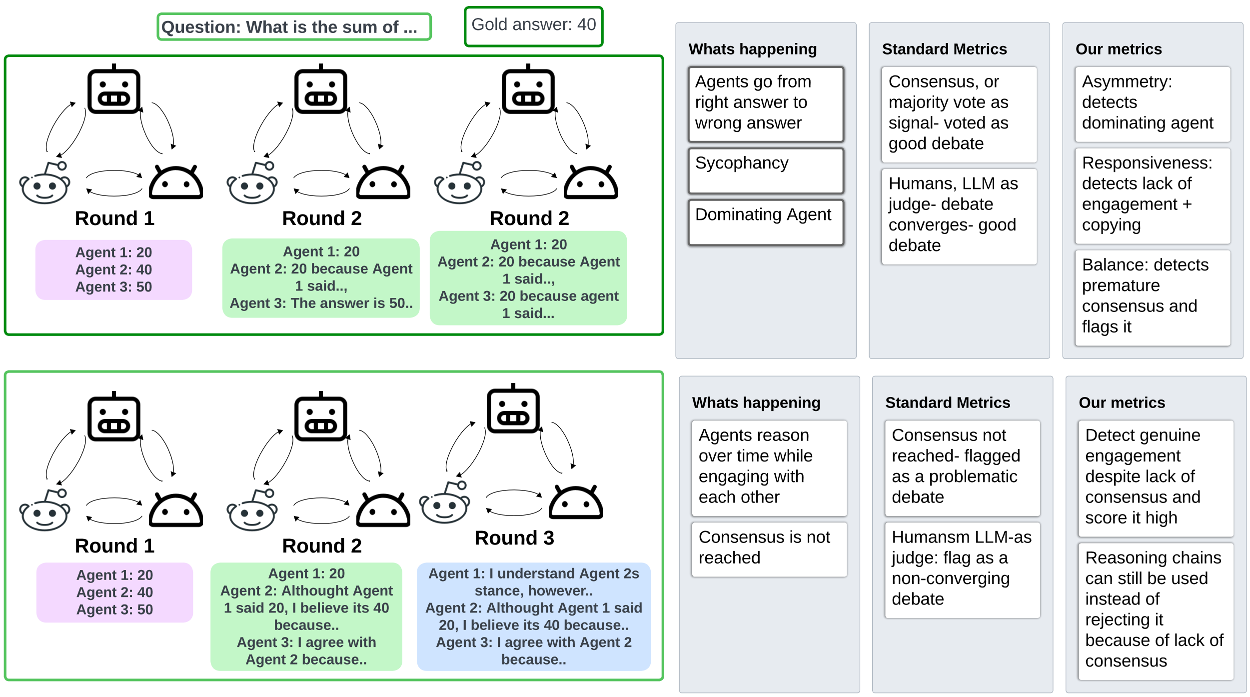


Figure 4. Intuition behind our diagnostic study

In objective datasets, wrong answers can be detected by comparing the generated answer against ground truth. When the answer is wrong, consensus is optimized via confidence weighed voting, prompt optimization. The root issue of the bad interaction is never addressed. In subjective settings with no ground truth, there is no way to know that this has happened because consensus and majority vote will signal those to appear correct.

In Debate 1, the agents reach a consensus using sycophancy and a dominating agent. The agent who had the correct answer changes it to an incorrect answer. However, standard metrics don't detect this- our metrics are able to. In Debate 2, agents don't reach a consensus, however, the thread of conversation is meaningful. Similarly, standard metrics do not detect this, ours do.

A.2. Dataset Details

We demonstrate our agent creation and multi-agent debate methods using both an objective benchmark and a real-world domains without ground truth.

For an objective dataset, we use MMLU dataset (Hendrycks et al., 2020).

For subjective datasets, we use city planning stakeholder interviews (Gosain et al., 2022). For city planning, we use 50 stakeholder interviews from (Gosain et al., 2022) with City of Miami stakeholders across four sectors—Nongovernmental Organizations (NGOs), Private Industry, Public Agencies, and Academia. Each participant completed one semi-structured interview with the same set of questions. This setting reflects real city planning processes, where multiple stakeholders must reach a single decision on housing, zoning, or infrastructure, despite there being no objectively correct answer (Innes & Booher, 1999).

For politics, we collect multiple interviews over time for current U.S. senators from CSPAN (C-SPAN, 2025) and major news outlets. We collect at least three interviews per senator, conducted over time and across different topics, capturing how their rhetoric, priorities, and stances evolve.

Table 4. Agent Sanity Check: Text similarity and quality metrics across agents on city planning and politics datasets.

Agent	Cosine Sim.	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore F1	MCQ Acc.
City Planning Dataset						
academic	0.60	0.22	0.02	0.11	**0.80**	0.95
public	0.67	0.18	0.03	0.11	**0.85**	1
private	0.65	0.12	0.02	0.09	**0.85**	1
ngo	0.59	0.18	0.02	0.10	**0.83**	1
Politics Dataset						
tim_scott	0.69	0.36	0.06	0.18	**0.86**	1
bill_cassidy	0.84	0.40	0.17	0.29	**0.91**	1
tammy_baldwin	0.78	0.32	0.11	0.21	**0.89**	0.90
bernie_sanders	0.76	0.30	0.13	0.20	**0.88**	1
kyrsten_sinema	0.63	0.22	0.03	0.13	**0.86**	1

We use interviews for our simulations because they capture rich information about interviewee beliefs, priorities, and reasoning style, and because interview-based agent construction is widely used in prior work on LLM social simulations (Park et al., 2023; 2024a). For each interviewee, we instruction-tune the model to act as that individual, using the interview questions and responses to condition the agent's behavior. We then evaluate each agent independently on held-out portions of the interview corpus, prompting the model with multiple-choice questions generated from unseen interview segments to test generalization beyond source material.

For debate experiments, we generate a large set of policy-style questions on which agents express their stances using five-point Likert ratings (1–5). Across multiple rounds, agents update these ratings and provide short justifications, producing structured behavioral trajectories used to compute our diagnostic metrics.

To externally validate our diagnostic metrics, we additionally evaluate on MMLU and BIG-bench Hard (BBH), where ground truth answers exist. These benchmarks allow us to test whether debates that score higher on our diagnostic measures also tend to achieve higher accuracy when correctness is well-defined. We use accuracy only as a validation signal in these settings, not as an optimization objective.

We will release our annotations and all test questions upon acceptance of the paper.

A.3. Implementation Details

A.3.1 Debate Instances and Number of Runs

All experiments use a fixed multi-agent debate protocol with $A = 3$ agents and $T = 5$ interaction rounds.

Objective dataset (MMLU). We randomly sample 100 questions from the MMLU benchmark. For each question and model, we run 5 independent debate instances with different random seeds, resulting in 500 debates per model. All reported diagnostic scores, correlations, and accuracies are averaged across runs.

Subjective datasets. For the city planning dataset, we generate 50 policy-style questions, each debated by three stakeholder agents. For the politics dataset, we generate 40 policy-style questions, each debated by three political agents. Each question is evaluated using three independent debate runs, yielding 150 debates for city planning and 120 debates for politics per model.

Agent ordering is randomized at the start of each debate. Question sets, agent personas, and prompts are held fixed across runs; only the random seed is varied to isolate stochastic generation effects.

A.3.2 Model Inference Settings and Randomness

All models are evaluated using identical inference settings. We use temperature sampling with $\tau = 0.7$ and nucleus sampling with top-$p = 0.9$. No top-k truncation is applied.

Each debate run is assigned a fixed random seed controlling agent generation and turn ordering. Seeds are varied across runs but shared across models for matched comparisons. No model-specific prompt tuning or decoding optimization is performed.

A.3.3 Agent Construction

Agents are constructed via prompt-based persona conditioning rather than parameter fine-tuning.

City planning agents. We use 50 stakeholder interviews from Gosain et al. (2022), spanning four sectors: non-governmental organizations (NGOs), private industry, public agencies, and academia. For each stakeholder, we construct a system prompt containing (1) a role description (e.g., "You are a city planning stakeholder representing a public agency in Miami") and (2) a summarized version of the interview responses describing priorities, constraints, and reasoning style.

Politics agents. For each U.S. senator, we collect at least three interviews conducted across different topics and time periods. These interviews are concatenated into a persona prompt describing the senator's policy positions, rhetoric, and reasoning tendencies.

When interview transcripts exceed the model context window, we retain the most recent and policy-relevant segments. Persona prompts are fixed across all runs and models.

A.3.4 Debate Prompting Protocol

At the start of each debate, agents receive the policy question, their persona prompt, and instructions to state an initial position with a brief justification.

In subsequent rounds, agents are shown anonymized messages from the other agents and instructed to consider opposing arguments, update their stance if warranted, and provide a short justification. Agents are not informed of any diagnostic metrics, outcome measures, or evaluation criteria.

A.3.5 Stance Representation and Extraction

At each round, agents are explicitly instructed to report their stance using a five-point Likert scale: (1) Strongly Disagree, (2) Disagree, (3) Neutral, (4) Agree, and (5) Strongly Agree. These values are mapped linearly to numerical stances $x_a^t \in \{-2, -1, 0, 1, 2\}$.

Prompts enforce a structured response format requiring the numeric stance to appear on a separate line. If an agent fails to produce a valid stance value, it is re-prompted once. Debates with unresolved parsing failures are discarded, accounting for less than 1% of all runs.

A.3.6 Reasoning Quality Scoring

Each agent message is assigned a reasoning quality score $q_a^t \in [0, 1]$ using the RECEVAL framework (Prasad et al., 2023).

The evaluator model is applied independently to each agent turn and observes only the agent's current justification text. It does not observe other agents' messages, final debate outcomes, or ground-truth answers (when available), ensuring that reasoning quality reflects justification coherence rather than correctness.

A.3.7 Moderator Protocol

If consensus is not reached after the final debate round, a moderator agent is introduced. The moderator is instantiated using the same base model as the debating agents and is shown the full debate transcript with anonymized agent identifiers.

The moderator is instructed to select a final answer based on the arguments presented. The moderator's decision is used only for outcome-level evaluation. All process-level diagnostic metrics are computed exclusively from agent interactions and do not depend on the moderator's output.

A.3.8 Human Preference Evaluation

Human evaluation is conducted for subjective datasets using pairwise preference judgments. We randomly sample 60 debate pairs per dataset, where each pair consists of two debates generated from the same question under different experimental conditions.

Annotators receive the following instructions:

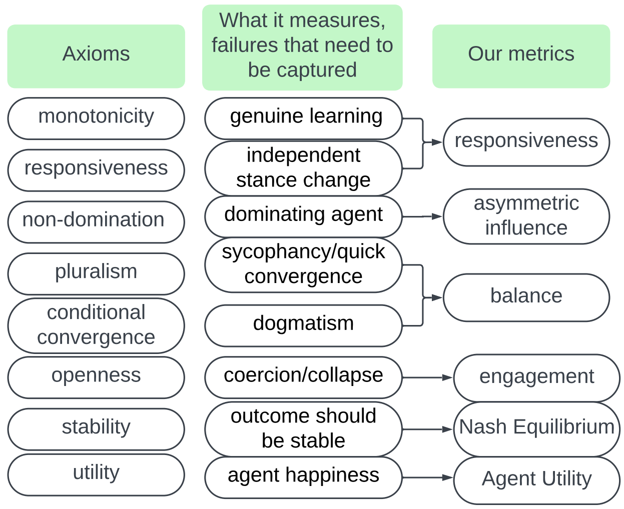


Figure 5. Axioms to metrics mapping and its motivation

Table 5. Pairwise accuracy with human preferences on political debate simulations across three model families. Values indicate the fraction of debate pairs for which the metric correctly predicts the debate preferred by human annotators (chance = 0.5).

Signal / Metric	**GPT-4o**	**Qwen2.5**	**Gemini-1.5**
Num. tokens	0.50	0.49	0.50
Rounds	0.50	0.49	0.50
LLM-as-Judge	0.61	0.58	0.60
Nash-stability	0.62	0.60	0.61
Agent Utility	0.64	0.61	0.63
Engagement	0.68	0.65	0.67
Responsiveness	0.74	0.71	0.73
Balance	0.73	0.70	0.72
Influence Asymm. (inv.)	0.69	0.66	0.68
All Process Metrics	**0.76**	**0.73**	**0.75**

> “You will be shown two debate transcripts. Your task is to select the debate that exhibits higher-quality deliberation, considering factors such as engagement, responsiveness, fairness of influence, and coherence of interaction. Do not judge based on agreement or which side you personally prefer.”

Annotators are blinded to model identity, metric values, and experimental conditions. Each debate pair is evaluated by three independent annotators with backgrounds in communication or social sciences. Final labels are obtained via majority vote. Inter-annotator agreement is measured using Cohen’s κ.

A.4. Axiom to Metric Mapping

Figure 5 shows axiom to metric mapping and its motivation.

B. Politics Dataset Results

Table 5 shows the results for the human preference correlation test for the politics dataset.